

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CSC 223 — Past Questions & Materials
by Renance

Course: CSC 223
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 3
Levels: 200

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Paper 1 of 3 — CSC 223 DISCRETE STRUCTURE 2022, 2023 Past Questions.

Level 200 • Second Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF SCIENCE

DEPARTMENT OF COMPUTER SCIENCE

SECOND SEMESTER EXAMINATION 2022/2023

Course Code: CSC223°

Title: DISCRETE STRUCTURE

Instructions: Attempt any 4 questions. Time Allowed: 2Hours.

Question 1

a) A graph in which the edges are one-way are called... Directed graph also Digraph

Assignment of colours to the vertices of a graph G is Vertex coloring

b) A graph

if there exist a one-one correspondence between them

A graph with only one vertex and no edges is G

A closed path that visits every vertex in G exactly once

c) Two graphs are... isomorphic

A vertex is called Isolated Vertex

d) A vertex in a graph G containing no edge

A graph is Connected if there is a path between any two vertices

f) A graph is connected

p) An assertion is

is called.....

q). An edge in a graph

any two of its vertices

e) A graph is

A graph G is called a Cutpoint vertex if $G - v$ is disconnected

g) A relation R is.....Transitive....if whenever aRb and bRc , then aRc .

h) A vertex of degree zero is called Isolated vertex

A graph in which all edges are distinct is called a Trail

i) A path in a multigraph in which

j) A graph is called trivial graph...

k) The acronym DFS in graph theory is Depth First search...

A graph algorithm means

1) A path in a multigraph in which all vertices are distinct is called Simple path...

Tree..if it is connected and has no cycles

m) A graph is called a

nce is called

n) A circuit in a graph G which is closed

Hamiltonian

if and only if each vertex has even degree

o) A finite connected graph is Eulerian

(premises) yields another proposition (conclusion)

such that a given set of propositions

Argument

path with both endpoints at v is called

.Breadth first search

t) The acronym BFS in graph algorithm means

partitioned into two subsets M and N

Symmetric... If whenever aRb then bRa where $a, b \in A$

t) A relation R on a set A is

of a sequence of events where each

u) Tree Diagram is used to enumerate all the possible outcomes

event can occur in a finite number of ways.

w) The set of all ordered pairs (a, b) where $a \in \text{set } A$ and $b \in \text{set } B$ is called Product set -

s) A graph G is said to be bipartite if its vertices V can be partitioned

such that each edge of G connects a vertex of M to a vertex of N

Recess Bridge if $G - e$ is disconnected.

y) A graph G is said to be... Complete, if every vertex in G is connected to

z) The number of edges that lead to a vertex v can be termed the... Degree of a vertex of the vertex.

Weighted Graph if each edge of the graph is "

600300000 Regular if every vertex

ff) A subgraph T of a connected graph G is called a... Spanning tree

vertices of G gg) Let u and v be vertices of a graph. If $u = v$ then the edge that connects

a... Adjacent hh) In a digraph, the number of edges beginning at a vertex v is the

x) An edge e in a connected graph G is called a

aa) A graph $G = G(V, E)$ and another graph $H = H(V', E')$, H is a... subgraph if $V_A \subset V$ and $E' \subset E$. bb)

Spanning of G if T is a tree and T includes all the

v. ii) The number of edges ending at a vertex v in a digraph is called the...

every other vertex in G

v) The principle that two different events can occur in different number of ways but cannot occur simultaneously is called the... The relativity of simultaneity.

The length of the shortest path between two vertices u and v in a graph is called..... Diameter $cc)$

A

graph is called a..... Labeled Graph, if its edges and/or vertices are assigned data of one kind or another :

dd) A graph is called a.....- assigned a nonnegative number. ee) A graph is in the graph has the same degree.

cts them is called

Se sd Out Degree of

In Degree of v . jj) Let

G be a digraph, a vertex v with no edge beginning from it is called..... kk) Let G be a digraph, a vertex v with no edge ending at it is called a :

bees! if for any pair of vertices u and v in G , there is a path from u to v

mm) A digraph G is..... Strongl

vertices u and v in G , there

nn) A relation R on a set A is.....-

II) A digraph G is Strongly Connected

and a path from v to u .

y Connected is a path from u to v or a path from v to u . if for any pair of

Reflexive, if aRa for every $a \in A$

Paper 2 of 3 — CSC 223 Requirements engineering Handout

Level 200 • Second Semester • Handout

INTRODUCTION

Requirements engineering, widely recognized as the first phase of software engineering process, is considered the key task of software development (Wahono 2003; Asghar & Umar 2010).

Ambiguous requirement is reported as one of the top reasons for software project failure (Hoffman & Lehner 2001) and product defects

(Young 2001). Indeed, an effective requirements engineering is essential to the rest of software development process.

Requirements engineering is also a crucial factor which influences productivity and product quality

(Fernandez et al 2011).

Poor Requirement gathering is one of the primary causes of project failure. Studies (such as those by the standish group CHAOS report) have shown that incomplete, inconsistent or poorly managed requirements significantly increase the risk of cost overruns, project delays and unsatisfactory outcomes.

In order to develop high software systems that is user friendly and meet stakeholders' satisfaction, while trying to avoid traditional requirement engineering problems, an effective requirement engineering process along with appropriate guidelines is inevitably required.

DEFINITIONS

Requirement engineering is defined as a branch of software engineering concerned with the process defining documenting and maintaining the requirement in the engineering design process.

It can also be defined as a process of formalising and defining needs at each stage of system designing including customer requirements, design requirement and testing requirement, it is the process of defining, eliciting, analysing, specifying validation.

Requirement engineering is an addition to software engineering; it focuses on the essential concepts the implementation of system engineering.

REQUIREMENT ENGINEERING PROCESSES

A requirements engineering phase consists of five core activities;

□ Requirements Elicitation: The process of gathering information from stakeholders, users, domain experts, and other sources to understand their needs, goals, and

expectations for the system.

□ Requirements Analysis: The process of examining, refining, and prioritizing gathered requirements to ensure they are clear, consistent, feasible, and free from conflicts or ambiguities.

□ Requirements Specification: The process of documenting the agreed-upon requirements in a structured, precise, and testable format, often in a Software Requirements Specification (SRS) document

□ Requirements Validation: The process of confirming that documented requirements accurately reflect stakeholder needs and are complete, correct, and feasible for implementation.

□ Requirements Management: The process of tracking, controlling, and maintaining requirements throughout the project lifecycle, ensuring that any changes are systematically evaluated and implemented.

TYPES OF REQUIREMENT ENGINEERING

Business Requirement: High-level goals and objectives of the organization or project

Stakeholder Requirement: this is the needs and expectations of the stakeholder (user, manager, customer etc.)

Functional Requirement: these are specific features or functions that the system must perform.

Non-functional Requirement: this are the constraint and quality attributes the system must satisfy.

System Requirement: These are detailed descriptions of the systems' capabilities, combining functional and non-functional requirement.

SOFTWARE DEVELOPMENT LIFE CYCLE

(SDLC)

SDLC is a process followed for software building within a software organization. SDLC consists of a precise plan that describes how to develop, maintain, replace, and enhance specific software. The life cycle defines a method for improving the quality of software and the all-around development process.

STAGES OF THE SOFTWARE DEVELOPMENT LIFE CYCLE

SDLC specifies the tasks to be performed at various stages by a software engineer or developer. It ensures that the end product is able to meet the customer's expectations and fits within the overall

budget. Hence, it's vital for a software developer to have prior knowledge of this software development process. SDLC is a collection of these six stages, and the stages of SDLC are as follows:

STAGE 1: Planning and Requirement Analysis

STAGE 2: Defining Requirements

STAGE 3: Designing Architecture

STAGE 4: Developing Product

STAGE 5: Product Testing and Integration

STAGE 6: Deployment and Maintenance of Products

SOFTWARE DEVELOPMENT LIFE CYCLE

MODELS

Software Development Models are structured frameworks that guide the planning, execution, and delivery of software projects. They define the sequence of development stages, such as requirements, design, coding, testing, and deployment.

Waterfall Model

- o Agile Model

- o V-Model

- o Spiral Model

- o Incremental Model

- o RAD Model

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